GRAND ROUNDS CASE · FOLLOW-UP PROGRAM

Followup 26-17

Mixed sedative and opioid overdose with recurrent apnea

Patient	A 64-year-old male with hypertension, depression, alcohol use disorder, prior pericarditis, charted paroxysmal atrial fibrillation, and chronic back pain treated with oxycodone-acetaminophen. His family described months of cognitive decline and a possible recent stroke. **He was found unresponsive after ingesting quetiapine, temazepam, cyclobenzaprine (Flexeril) and oxycodone-acetaminophen, after heavy recent drinking.**
Outcome	Survived. Intubated in the emergency department for airway protection and extubated on hospital day 2. **His QRS measured 162 ms with a right bundle branch block pattern that resolved as the drugs cleared.** He was also hypothermic at 34.1 °C, had several acute left rib fractures, and on MRI had a small new right frontal stroke and an older one.
Disposition	Admitted to intensive care, then the medical floor. Petitioned and certified for psychiatric treatment. On hospital day 7, at the time of these records, his disposition was pending between inpatient psychiatry and inpatient rehabilitation.

The call

The call was dispatched as a possible overdose. En route, the crew was told it was a cardiac arrest with CPR in progress; his family had started compressions on the dispatcher's instructions. A second update said he was unresponsive but breathing after police gave naloxone 4 mg intranasally. Police reported slow, shallow respirations that improved with the naloxone.

The crew found him undressed on the bathroom floor, lying on his left side, unresponsive with a GCS of 4, breathing on his own. His wife had found empty bottles of quetiapine, temazepam and cyclobenzaprine and a missing bottle of his oxycodone-acetaminophen, all of which had pills in them earlier that day. The crew found a firm abdominal mass and bruising on the left foot. They applied a cervical collar, obtained a 12 lead showing a right bundle branch block, placed two 16 gauge IVs and started saline.

An attempted nasopharyngeal airway made him cough and gag, and it was removed. During transport he stopped breathing twice. Each time the crew ventilated him by bag-valve-mask and gave naloxone 2 mg IV, and each time he resumed breathing on his own. He kept his gag reflex throughout, needed suctioning, and responded only by grimacing to pain.

Times below are from patient contact.

From contact	Finding or action
Before contact	Naloxone 4 mg intranasal by police
+2 min	BP 129/74, pulse 74, respirations 14, SpO2 98%, glucose 169, GCS 4
+6 min	12 lead: right bundle branch block
+12 to +16 min	Two 16 gauge IVs; normal saline

From contact	Finding or action
+14 min	BP 184/100
+17 min	Nasopharyngeal airway attempted; gag and cough; removed
+23 min	Leaves scene
+24 min	End-tidal CO2 20 mmHg
+28 min	Apneic; bag-valve-mask ventilation
+30 min	Naloxone 2 mg IV; breathing resumes
+38 min	Apneic again; bag-valve-mask ventilation
+39 min	Naloxone 2 mg IV; breathing resumes
+44 min	End-tidal CO2 35 mmHg
+48 min	Arrives at the emergency department

The numbers to hold onto are a GCS of 4 that never changed with naloxone, two episodes of apnea that did, and a QRS of 162 ms.

Primary impression

The crew's impression was drug overdose. The hospital's lead diagnosis was acute respiratory failure with hypoxia and hypercapnia from the ingestion. This crew recognized which part of a mixed overdose naloxone could treat. They did not chase his level of consciousness with escalating doses. They watched his breathing, ventilated him when it stopped, gave naloxone to bring it back, and trended his end-tidal CO2 through transport. That is the right management of a mixed ingestion.

The mechanism underneath is several drugs acting on different receptors at once. The opioid suppressed his drive to breathe and responded to naloxone. Temazepam, quetiapine, cyclobenzaprine and alcohol suppressed his consciousness through receptors naloxone does not touch.

Hospital course

Emergency department, day 0. He arrived breathing, with pinpoint pupils, coarse breath sounds, slow respirations with occasional gasps, and a GCS of 5. He was intubated for airway protection by rapid sequence intubation with etomidate and rocuronium, on the first attempt with video laryngoscopy. His first arterial gas showed a pH of 7.28 and a PCO2 of 58. His temperature was 34.1 °C, and active warming was started. Poison control recommended activated charcoal through the gastric tube once the airway was protected, an acetaminophen level at four hours from ingestion, and keeping magnesium above 2.1 and calcium above 9.5.

His ECG showed sinus bradycardia at 57, a QRS of 162 ms, a right bundle branch block, and a QTc of 490. ST elevation appeared in V5 and V6 on the monitor. Cardiology read it as a right bundle branch block or a nonspecific conduction delay, not an infarction. His troponin went from 36 to 62 to 63 and then fell to 46. CT of the head was normal. CT of the cervical spine showed no fracture but severe degenerative stenosis. CT of the chest, abdomen and pelvis showed several acute nondisplaced left rib fractures and opacities at both lung bases, at least partly atelectasis but possibly aspiration. He was admitted to intensive care on propofol.

Hospital day 1. The widened QRS and the right bundle branch block resolved. MRI showed a small acute or subacute infarct in the right inferior frontal gyrus and an older infarct in the right corona radiata that had not been known. Neurology saw him and recommended outpatient cardiology follow-up about anticoagulation, given his charted atrial fibrillation.

Hospital day 2. He passed a spontaneous breathing trial and was extubated to nasal cannula. He remained confused and was moved out of intensive care.

Days 3 through 7. His confusion cleared over several days, and he was walking 800 feet with a walker. His kidney function returned to normal. His home aspirin and clopidogrel were continued. He was petitioned and certified for psychiatric treatment, and at the time of these records his disposition was pending between inpatient psychiatry and inpatient rehabilitation.

Final diagnoses: Acute respiratory failure with hypoxia and hypercapnia. Overdose. Acute encephalopathy. Acute ischemic stroke. Acute kidney injury. Macrocytic anemia. Alcohol use disorder. Closed fractures of multiple left ribs.

Education points

Naloxone in a mixed ingestion

Let's start with what each drug did. Opioids act on mu receptors in the brainstem respiratory centers and reduce the drive to breathe: the rate slows, breaths get shallow, and breathing eventually stops. Naloxone displaces the opioid from those receptors and restores the drive. It does nothing else.

Temazepam is a benzodiazepine and acts on GABA-A receptors, as alcohol does. Quetiapine blocks histamine, alpha-1 and muscarinic receptors. Cyclobenzaprine, sold as Flexeril, is built on the same three-ring structure as the tricyclic antidepressants and shares their sedating and anticholinergic effects. All of these depress consciousness, and none of them respond to naloxone. Benzodiazepines and opioids together also depress breathing more than either alone.

That is why his GCS stayed at 4 through 8 mg of naloxone while his breathing came back with each dose. The goal of naloxone in a mixed ingestion is adequate ventilation, not wakefulness. Escalating doses to wake a patient whose sedation comes from other drugs adds nothing, and in an opioid-dependent patient it can precipitate withdrawal.

Naloxone also wears off before most opioids do. Its effect lasts roughly 20 to 90 minutes, depending on the dose relative to the opioid on board. Oxycodone acts for hours, and a large ingestion keeps absorbing from the gut. When naloxone wears off, the opioid takes the receptors back and breathing slows again. This is called renarcotization, and it is what happened to him twice in transport.

Oxycodone and hydrocodone are both full mu agonists with similar durations, and naloxone reverses either the same way. The exceptions are methadone, which lasts far longer than any naloxone dose, and buprenorphine, which binds so tightly that standard naloxone doses may not displace it.

Sodium-channel blockade and the QRS

Cyclobenzaprine (Flexeril) shares more than structure with the tricyclic antidepressants. In overdose it can block the fast sodium channels that carry the rapid upstroke of the cardiac action potential. When those channels are blocked, depolarization spreads more slowly through the ventricles and the QRS widens. The right bundle is the most sensitive to that blockade, so the widening often looks like a right bundle branch block, with a tall terminal R

wave in aVR. Quetiapine adds potassium-channel blockade and prolongs the QT; poison control's magnesium and calcium targets protect against that.

In tricyclic poisoning the QRS width predicts what comes next. A QRS over 100 ms predicts seizures, and over 160 ms predicts ventricular dysrhythmias. His measured 162. The standard treatment for a QRS widened by sodium-channel blockade is sodium bicarbonate, which supplies sodium to overcome the block and alkalinizes the blood. His widening resolved as the drugs cleared. Mild hypothermia also slows conduction and widens intervals, and his temperature of 34.1 °C contributed to the bradycardia.

The ST elevation in V5 and V6 is a secondary change. When the QRS widens, repolarization changes with it, and the ST segments move away from the direction of the terminal QRS. Cardiology read it that way, and his small troponin rise fits a patient who was hypoxic, cold and found down rather than an infarction.

> **Further reading on sodium-channel blockade on the ECG.** The LITFL ECG library page on tricyclic overdose, with example tracings of the wide QRS, the terminal R wave in aVR, and the QRS thresholds. https://litfl.com/tricyclic-overdose-sodium-channel-blocker-toxicity/
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> **Further reading on opioid intoxication.** The EMCrit Internet Book of Critical Care chapter on opioid toxicity, including how naloxone's duration compares with the opioid it reverses and how to dose it to ventilation. https://emcrit.org/ibcc/opioid/

Found down

A patient found down after an overdose carries more than the poison. He was undressed on a tile floor, and quetiapine and alcohol both dilate skin vessels and impair temperature regulation. His temperature was 34.1 °C on arrival. His creatine kinase was 680, from muscle compressed against the floor. His creatinine was 1.23 and normalized with fluids. He also had acute left rib fractures and severe cervical stenosis on CT.

The emergency department attributed his rib fractures to the family's compressions and documented redness over the sternum. The intensive care team considered recent falls. Either way, the family did the right thing. Dispatchers instruct compressions when they cannot tell whether a patient has a pulse, because compressions on a patient who does not need them cause injuries like these, while delayed compressions in a patient who does need them cause death.

Prehospital considerations

Naloxone treats breathing, not consciousness. Titrate it to an adequate rate and depth, with ventilation by bag-valve-mask in the meantime. A patient who stays unresponsive after the breathing returns is sedated by something naloxone does not reverse.

Apnea returns when the opioid outlasts the naloxone. Expect it within an hour, and watch for it with end-tidal CO2 and a respiratory rate rather than with the oximeter. This crew documented end-tidal values through transport and caught both episodes.

Every ingestion gets a 12 lead, and the QRS width is the number to report. A QRS over 100 ms after an overdose is sodium-channel blockade until proven otherwise. This crew's 12 lead captured the right bundle pattern six minutes after contact.

Breathing and protecting an airway are different endpoints. He breathed after naloxone, and he needed intubation for airway protection in the emergency department. Both are true. Position on the side, suction ready, and an airway plan for the patient who stays at a GCS of 4.

The interventions

PREHOSPITAL INTERVENTIONS

Naloxone 4 mg intranasal, by police before arrival. It restored his breathing, as police documented.

12 lead ECG. It recorded the right bundle branch block pattern six minutes after contact, the earliest record of the conduction effect.

Cervical collar. Applied for a patient found down with an unknown mechanism.

Two 16 gauge IVs and normal saline. Access for naloxone and fluid for a patient found down.

Nasopharyngeal airway, attempted. It triggered coughing and gagging and was removed. A patient who gags on an adjunct still has airway reflexes, and removing it was correct.

Bag-valve-mask ventilation and naloxone 2 mg IV, twice. Each apneic episode was treated by ventilating first and then restoring the drive to breathe, and each time he resumed breathing on his own.

Suction and end-tidal CO2 monitoring. End-tidal CO2 went from 20 to 35 mmHg across transport, which is a patient being ventilated and tracked rather than assumed.

IN-HOSPITAL INTERVENTIONS

Rapid sequence intubation. Etomidate and rocuronium, first attempt by video laryngoscopy, for airway protection in a patient with a GCS of 5.

Activated charcoal. Given through the gastric tube after intubation, once the airway was protected against aspiration.

Magnesium and calcium targets. Kept high-normal to stabilize repolarization against QT prolongation.

Active warming. A forced-air warming blanket and a temperature-sensing bladder catheter.

MRI and neurology consultation. Found the new and old strokes.

The labs, and what they mean

Reference ranges vary between institutions and analyzers. These are the ranges printed on the source reports.

Ventilation

Lab	Value	Reference range
pH, arterial	7.28	7.35 to 7.45
PCO2, arterial (mmHg)	58	35 to 48
PO2, arterial, on 100% oxygen (mmHg)	583	83 to 108

A PCO2 of 58 with a pH of 7.28 is an acute respiratory acidosis. His bicarbonate was a normal 24, so this was a new problem rather than a chronic one: hypoventilation from the opioid and sedatives, not yet corrected after intubation. His lactate was a normal 1.0, so his circulation was adequate on arrival.

Toxicology

Lab	Value	Reference range
Oxycodone screen, urine	Presumptive positive	Negative
Benzodiazepine screen, urine	Presumptive positive	Negative
Cannabinoid screen, urine	Presumptive positive	Negative

Acetaminophen and salicylate levels were normal, including the acetaminophen level at four hours. His ethanol was below 10, undetectable despite reported heavy drinking, which means the alcohol had cleared before the ingestion was found.

A standard opiate screen does not reliably detect oxycodone. His opiate screen was negative and his oxycodone screen positive. The opiate immunoassay is built to detect morphine and codeine. Oxycodone cross-reacts with it poorly, and synthetic opioids such as fentanyl not at all, so both need their own assays. A negative opiate screen does not rule out an opioid overdose.

Heart, muscle and blood count

Lab	Value	Reference range
hs troponin T, baseline (ng/L)	36	23 or less
hs troponin T, 2 hour (ng/L)	62	Less than 23
Creatine kinase (U/L)	680	20 to 200
Creatinine (mg/dL)	1.23	0.70 to 1.20
Hemoglobin (g/dL)	9.6	14.0 to 18.0
Mean cell volume (fL)	105.2	80.0 to 100.0

The troponin peaked at 63 and fell, consistent with injury from hypoxia, cold and being found down rather than a coronary occlusion. The creatine kinase and creatinine are muscle compression and dehydration, both corrected. His hemoglobin had been 12.1 three weeks earlier, and large red cells are the typical pattern of heavy alcohol use.

Glucose in the urine with a normal blood glucose is the signature of an SGLT2 inhibitor. His urine glucose was over 1,000 mg/dL against a blood glucose of 104 and a printed range of negative. He was prescribed dapagliflozin, which works by making the kidney spill glucose. The finding is expected, and it identifies a drug class that can cause ketoacidosis with a near-normal glucose in a patient who is fasting or drinking heavily. His bicarbonate and anion gap were normal.

Learning points

1. **In a mixed ingestion, naloxone restores breathing, not consciousness.** His GCS stayed at 4 through 8 mg of naloxone, because the benzodiazepine, quetiapine, cyclobenzaprine and alcohol act on receptors naloxone does not reach. Dose to ventilation.

2. **When the opioid outlasts the naloxone, apnea comes back.** Naloxone lasts roughly 20 to 90 minutes, and oxycodone lasts hours. He stopped breathing twice in transport, and the crew ventilated and re-dosed each time.
3. **A wide QRS after an ingestion is sodium-channel blockade until proven otherwise.** Cyclobenzaprine (Flexeril) is a tricyclic in structure. His QRS was 162 ms, the threshold for ventricular dysrhythmias in tricyclic poisoning, and it resolved as the drugs cleared.
4. **Breathing and protecting an airway are different endpoints.** He breathed on his own after naloxone and still needed intubation for airway protection. Both the crew and the emergency department described him correctly.
5. **Found down adds cold, muscle injury and trauma to any poisoning.** His temperature was 34.1 °C, his creatine kinase 680, and he had acute rib fractures. Check a temperature, look for pressure injury, and protect the spine when the mechanism is unknown.

Prepared from records obtained through the follow-up request process and de-identified for education. To request follow-up on a call you ran, fill out the request form at https://www.berrienmca.org/followup/